Harshil Agrawal

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EDUCATION

Virginia Tech | M.S. Data Science | Blacksburg, VA | Aug 2026 to May 2028

SVKM's NMIMS University, MPSTME, Mumbai | B.Tech Computer Engineering, GPA: 3.60/4.0 | Graduated May 2026

EXPERIENCE

Prowess Consulting | Data & AI Engineering Intern | Dec 2025 - Jun 2026

- Shipped a WhatsApp ordering assistant for pharmaceutical distribution on Azure App Service that has handled 10,000+ orders from about 1,000 pharmacies, using Azure OpenAI intent classification over a 1,500+ product catalog with Snowflake-backed conversation logs.
- Built SalesVitals, a FastAPI customer-insights tool that shows pharmaceutical field reps a customer's full history before each call; before it, every outreach was manual, with no view of the customer.
- Removed per-request Snowflake queries and their recurring cost by serving SalesVitals from a scheduled SQLite snapshot, validated on every build in GitHub Actions.
- Maintained reproducibility with Git, GitHub Actions, and automated test suites.

Gram Panchayat, Maharashtra | Social Impact Data Analyst, Volunteer | 2024

- Collected and analyzed community participation, sanitation, and education data for rural development.
- Tracked 500+ sapling plantations and evaluated sanitation and digital-literacy programs.
- Worked with local authorities on resource allocation for old-age homes and orphanages.

SELECTED PROJECTS

Mortgage PII Service | Python, FastAPI, spaCy, Presidio, Docker | 2026

github.com/Harshil411/banking_PII

- Built a Dockerized NLP service in FastAPI that finds and redacts 19 types of personal data in US mortgage and loan-servicing documents, from SSNs and routing numbers to borrower names, and runs with networking disabled so sensitive text stays on the machine.
- Resolved overlapping matches by strength of evidence, letting checksum validators (SSA, ABA routing, Luhn) outrank spaCy NER confidence, and returned every rejected match with its reason so each redaction can be audited.
- Reached 0.977 strict micro-F1 (0.981 precision, 0.974 recall) on 2,070 labeled entities in held-out synthetic documents; the evaluation and 267 pytest tests run on every push in GitHub Actions.

Manufacturing Defect Prediction | github.com/Harshil411/portfolio

- Deployed an inspectable Random Forest classifier through FastAPI. The project uses synthetic manufacturing sensor data and documents its model behavior and evaluation.

RESEARCH

A Hybrid LSTM + Deep Reinforcement Learning Framework for Enterprise-Grade Decision Support and Automated Trading

IEEE ICCICT 2026, accepted

- Proposes a hybrid LSTM and Deep RL framework that separates forecasting from execution.
- Validated on NIFTY-50 backtests with a Sharpe ratio of about 1.8 using LSTM, PPO, DQN, and SAC.

PROFESSIONAL DEVELOPMENT AND AWARD

Certificate of Participation in ACM SemiCode | Unstop | Mar 2024 | Credential ID: da0389db-6732-4f51-a3fb-4b97ec5dfe93

Workshops: Power BI Workshop; AI Prompting and N8N Automation Workshop, Prowess Consulting; AI and HPC Workshop, KLA and IIT Madras.

Award: Best Executive in Public Relations, Google Developer Student Clubs, NMIMS.

SKILLS

Programming: Python, SQL, Java, C, C++, JavaScript | Backend: FastAPI, Spring Boot, Spring Cloud, REST API | Cloud: Azure, Docker, GitHub Actions, CI/CD | Data and ML: Snowflake, SQLite, LSTM, Deep RL, scikit-learn, pandas